



3432 - A Unique Characterization of Early Quenching in a Young, Transitional Cluster at $z=1.84$ with NGDEEP and JEMS

Cycle: 2, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Stacey Alberts (PI)	University of Arizona
Dr. Nina Bonaventura (CoI)	University of Arizona
Dr. Christopher Nicholas Andrew Willmer (CoI)	University of Arizona
Dr. Eiichi Egami (CoI)	University of Arizona
Dr. Zhiyuan Ji (CoI)	University of Arizona
Dr. Katherine Suess (CoI)	Stanford University
Dr. Sandro Tacchella (CoI) (ESA Member)	University of Cambridge
Dr. Christina C Williams (CoI)	University of Arizona
Dr. Chris J. Willott (CoI) (CSA Member)	NRC Herzberg Institute of Astrophysics
Prof. Andrew Bunker (CoI) (ESA Member)	University of Oxford
Dr. Stephane Charlot (CoI) (ESA Member)	CNRS, Institut d'Astrophysique de Paris
Dr. Benjamin D. Johnson (CoI)	Harvard University
Dr. Jianwei Lyu (CoI)	University of Arizona
Lester Sandles (CoI) (ESA Member)	University of Cambridge
Mr. Fengwu Sun (CoI)	University of Arizona

OBSERVATIONS

ABSTRACT

Environment is a primary driver of galaxy evolution and yet it remains poorly understood even at its extremes in galaxy clusters. We propose a unique study using ultra-deep, archival NIRISS grism spectroscopy and NIRISS/NIRCam imaging to fully characterize the star forming, quenching,

and quenched populations in a young, transitional cluster at $z=1.84$, caught in the process of forming its passive population. Using the immediately public NGDEEP and JEMS surveys, we will disentangle the mechanisms driving early quenching as a function of the cluster accretion history by taking a full census of cluster galaxy properties (star formation histories, stellar ages, metallicities, size/morphologies, signs of merger activity, AGN) down to lower stellar masses ($\log M_{\text{stellar}}/M_{\text{sun}} \sim 7-8$) than ever achieved in clusters during cosmic noon. This point is critical: this study will enter the regime in which secular quenching in low-mass galaxies is rare and environmental quenching can be unambiguously identified. This study will set the benchmark for all future studies bridging the transitions from proto-clusters to quenched groups/clusters.

OBSERVING DESCRIPTION

This is an archival proposal to use the uniquely deep NIRISS spectroscopy from NGDEEP and NIRCам imaging from JEMS to fully characterize the populations in a galaxy cluster (designated HUDF46) at $z=1.84$ in the HUDF. HUDF46 is a transitional cluster, in the process of building its passive population (e.g. Red Sequence). NGDEEP will reach an emission line sensitivity of 10^{-18} erg/s/cm² in F115W, F150W, and F200W grism observations, reaching to $\log M/M_{\text{sun}} \sim 7-8$ in stellar mass, a population where environmental quenching dominates due to secular processes having long timescales. JEMS will provide high-resolution medium-band photometry at rest 0.6-1.7 μ m.

The identification of cluster members to low stellar mass will allow a $>2-3\times$ improvement in the current measurements of cluster properties (halo mass, velocity dispersion) and individual cluster member cluster-centric radii and line-of-sight velocities will be used to construct phase space diagrams that trace the accretion history of the cluster.

We will confirm the star forming and post-starburst cluster populations (to $\text{SFR} \sim 0.1 M_{\text{sun}}/\text{yr}$) using optical emission lines (H α + [NII], [OII], [OIII], H β), and characterize (ongoing and residual) SFRs, size/morphologies and merger activity, metallicities, stellar ages, etc using spectro-photometric SED fitting with non-parametric SFHs. The cluster populations will be compared to co-eval field populations selected from within the same datasets.

Quenching and quiescent galaxies will be confirmed using the Balmer/4000Å break and any detectable emission lines and similar properties will be measured and compared to field populations.

This complete look at the populations in a high redshift, transition cluster will reveal what drives early environmental quenching in massive halos, bridging the gap between proto-clusters and groups/clusters.